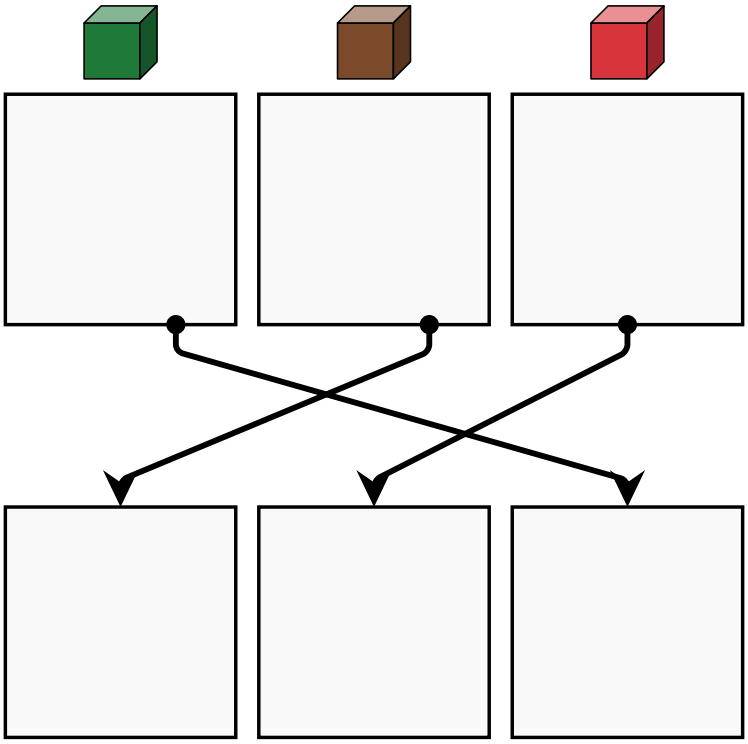
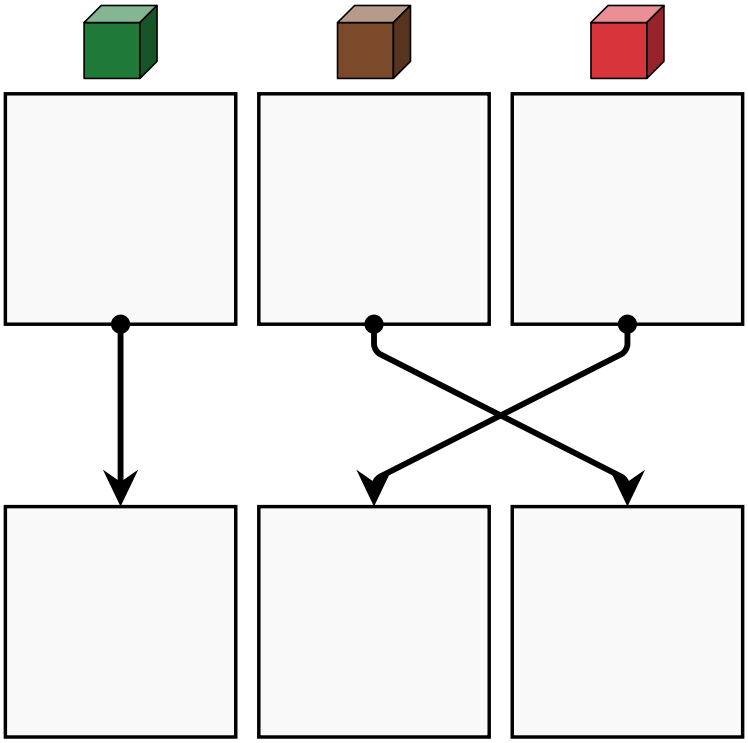
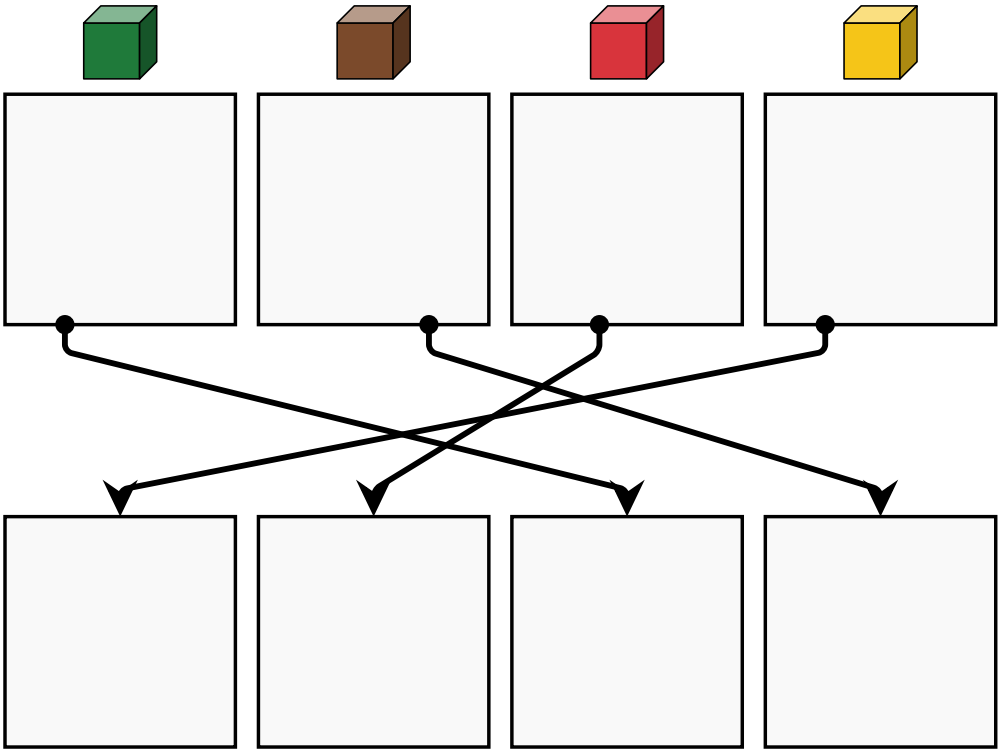
One turn: move every cube down its own arrow. Then slide the bottom row straight up into the top row. When you draw a machine, give every top slot one arrow going out and every bottom slot one arrow coming in.

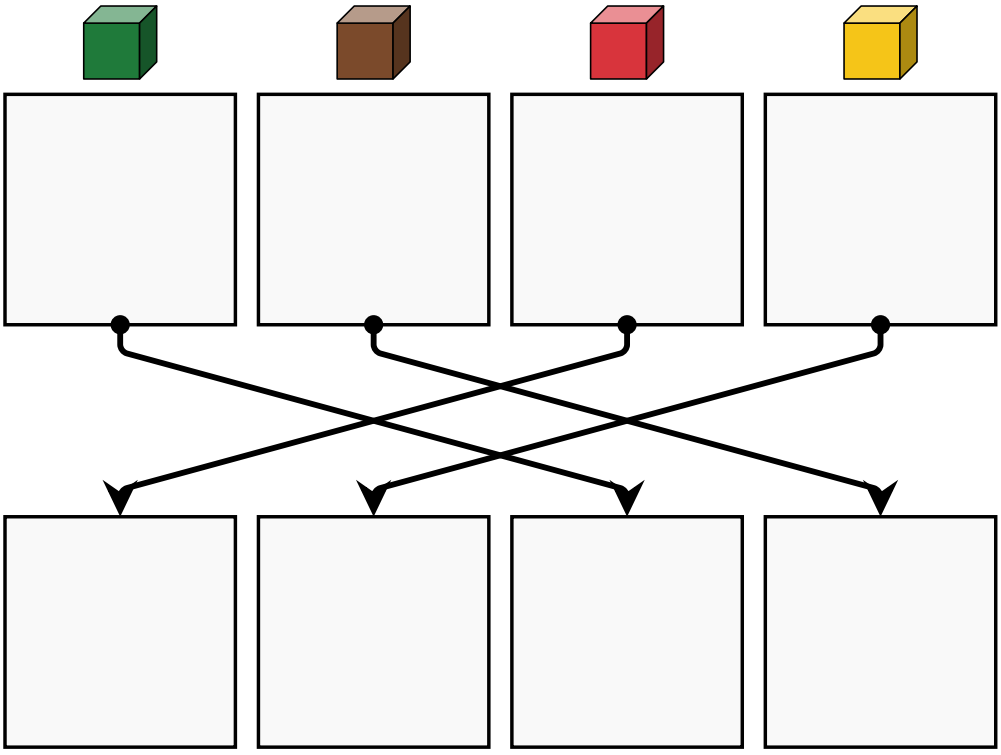
Problem 1: Put each cube in the top slot under its picture. Do turns until every cube is back under its own picture, and make a tally mark for each turn.



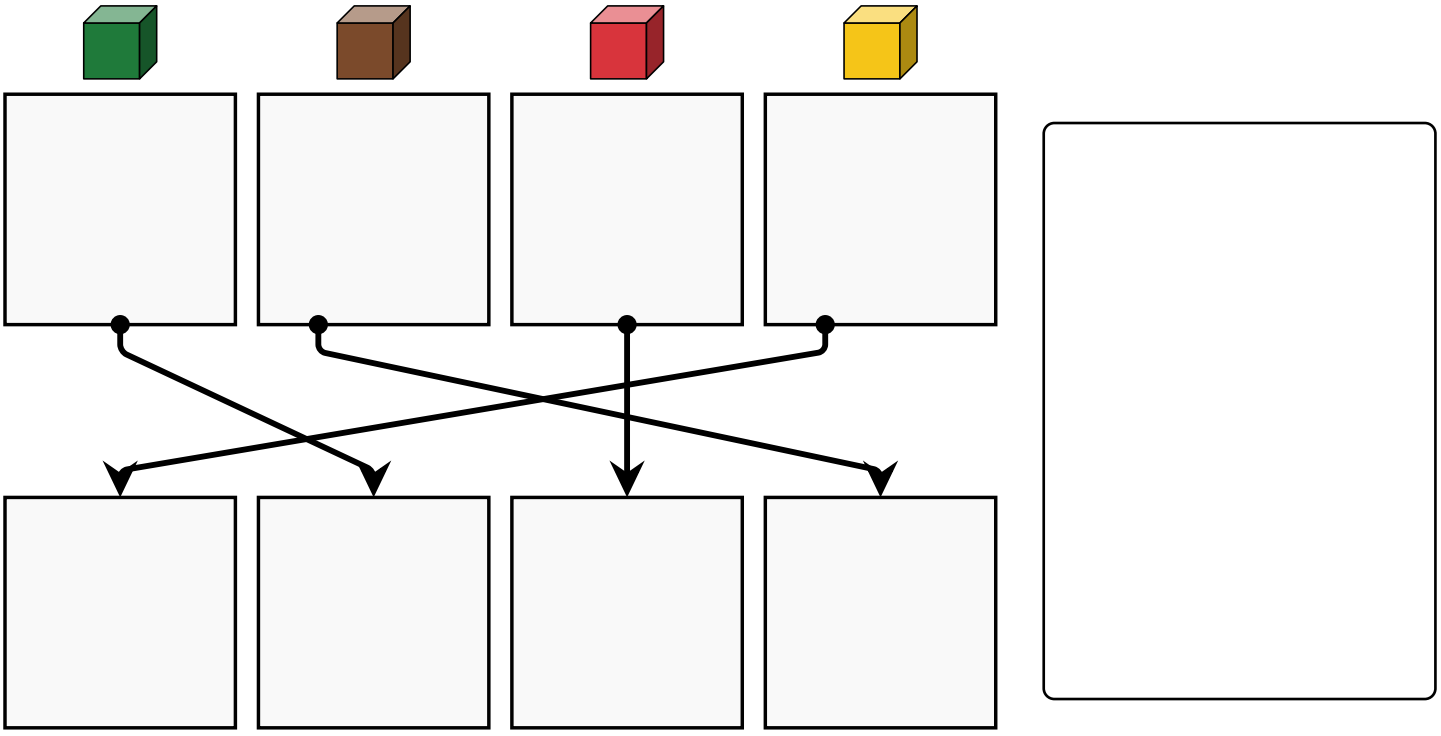


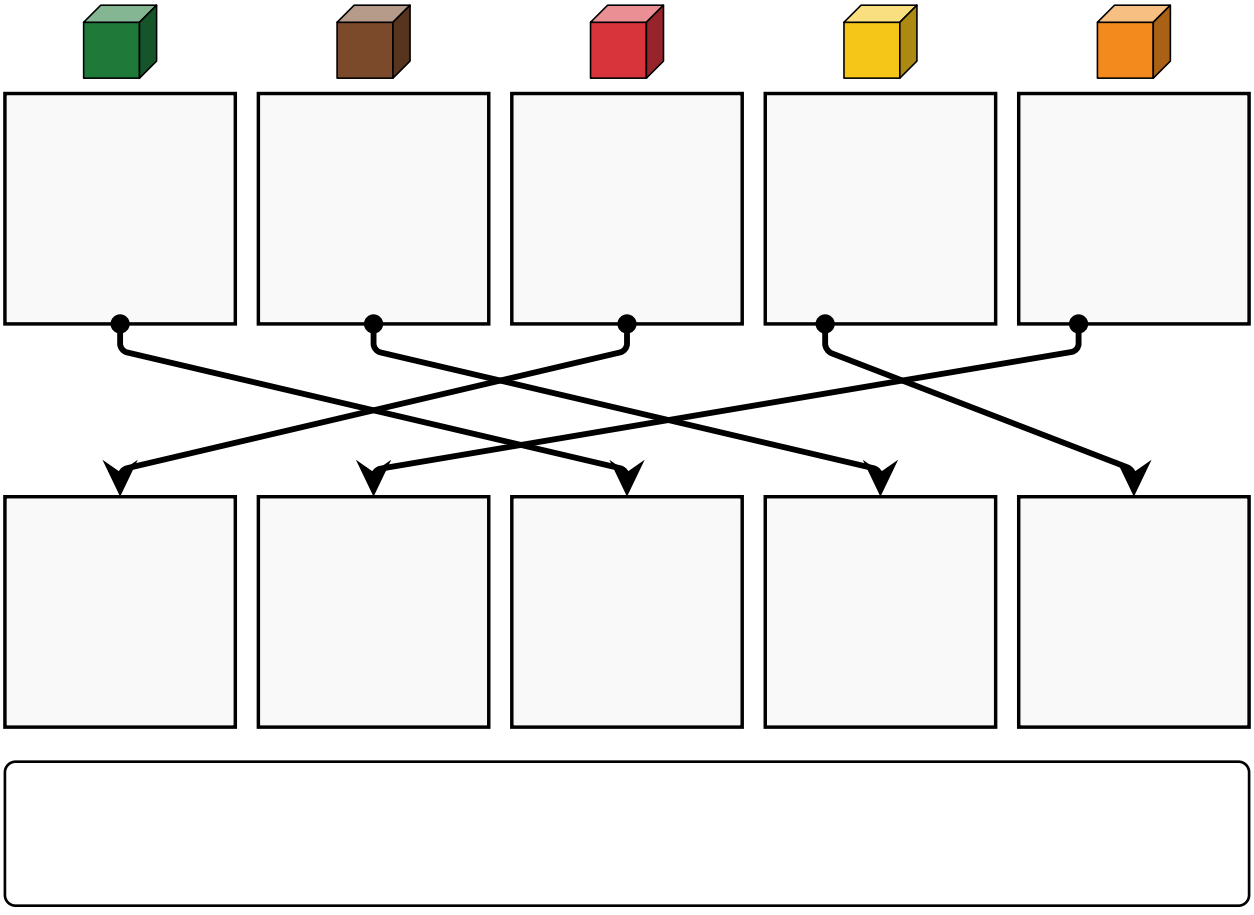
Problem 2: Do the same with four cubes on each of these machines.



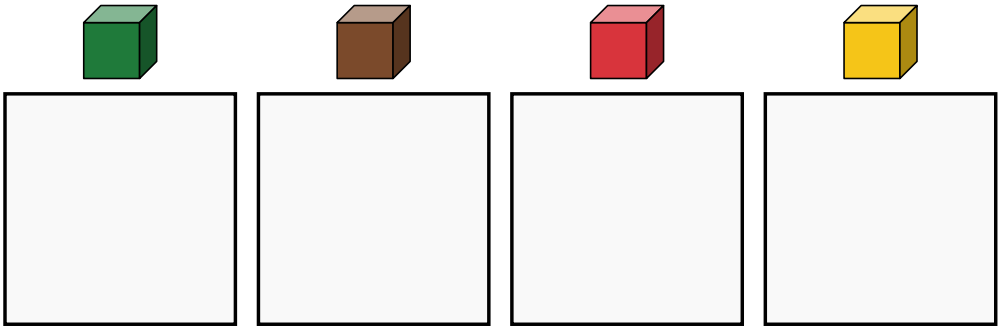


Problem 3: Do the same on these two machines.

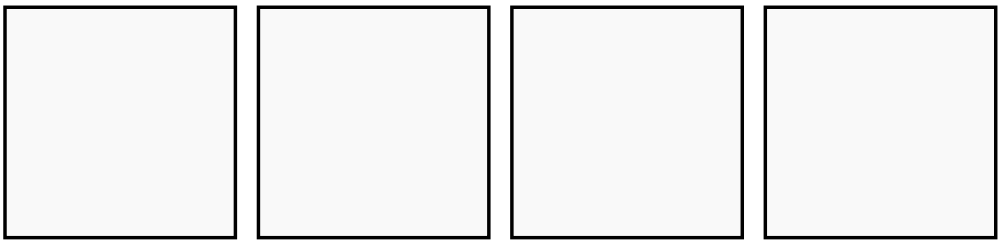


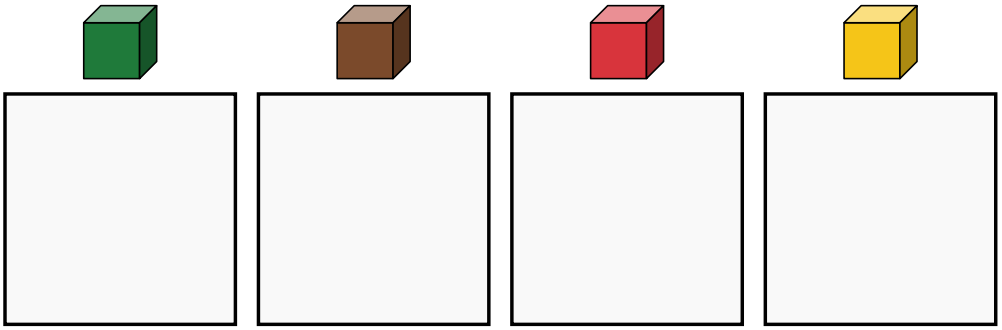


Problem 4: Draw arrows on each machine so the cubes get back under their pictures in the number of turns written beside it. Your partner checks each one with the cubes.

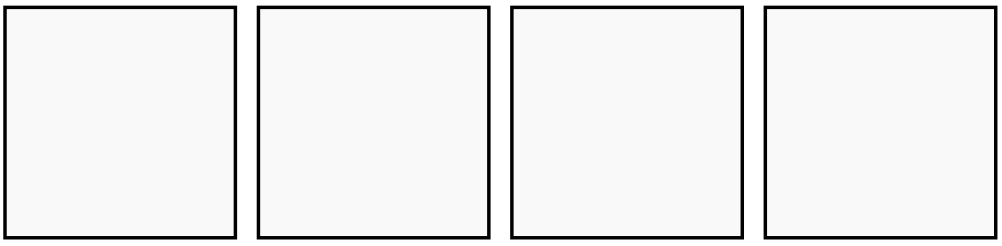


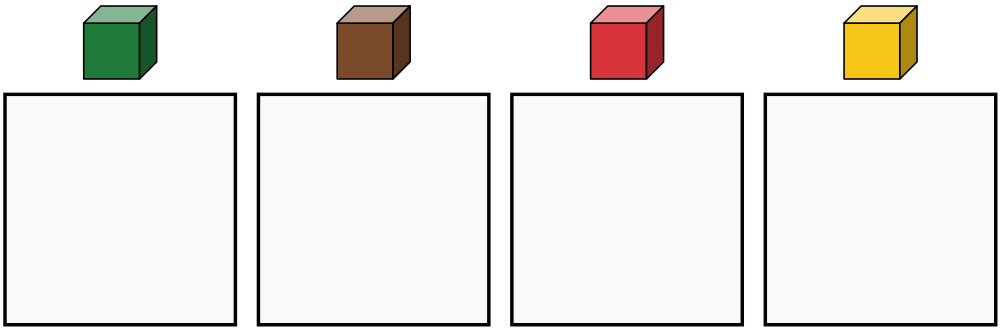
2 turns





3 turns





4 turns

